

# Can LLMs Do Causal Inference?

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## Abstract

This project investigates whether general-purpose, pre-trained large language models (LLMs) can perform credible causal inference via in-context learning (ICL). We design a structured evaluation framework using eight simulation engines that generate datasets with varying causal complexity. Simultaneously, We introduce four method-inducing ICL protocols, which serialize raw data into estimator-ready components to guide the LLM’s causal reasoning. In our controlled simulation environment, our findings indicate that these method-inducing protocols outperform raw-data prompting, helping the LLM estimator achieve an accuracy comparable to classical estimators like AIPW and regression adjustment in low causal effect settings. Notably, LLMs with ICL demonstrate superior robustness in settings with complex nonlinear confounding, maintaining a reliable 95% coverage where traditional methods fail. We then extend the framework to a real-world observational application using the Panel Study of Income Dynamics (PSID), where baseline covariates are measured in 2011, job-training participation is measured in 2013, and long-run labor income is measured in the 2023 wave. In this PSID analysis, both LLM-based protocols and classical causal estimators produce small and statistically insignificant estimates, with the LLM estimates broadly consistent with classical benchmarks and the AIPW-based protocol showing more cautious uncertainty quantification.

## 1 Introduction

With the rapid advancements of LLMs, their ability to summarize datasets, generate code, and produce statistical interpretations has made them increasingly popular as general-purpose data analysis assistants. As these capabilities expand, so does the temptation to rely on them to draw causal conclusions, even though these models were trained to focus on predicting text rather than performing causal inference. Estimating the average treatment effect (ATE) from observational data is fundamentally different from predictive modeling: causal estimation requires the explicit handling of confounding effects, overlap/positivity, nonlinearity, and uncertainty. Failure to consider any of these areas could potentially lead to biased or misleading causal conclusions.

ICL offers a practical way for steering a pre-trained LLM toward structured analytical tasks, without needing to modify its model parameters. By providing carefully designed prompts, which include clear explicit instructions, assumptions, output requirements, and few-shot examples, ICL attempts to better guide the model at inference time. In this project, we will evaluate whether ICLs can reliably induce LLMs to produce accurate and robust ATE estimates in both a controlled simulation setting and an observational analysis using data from the Panel Study of Income Dynamics (PSID).

## 2 Simulated Data Generation Framework

Evaluating causal inference methods on real-world data is difficult because of the fundamental problem of causal inference: we only see the outcome of the treatment a subject actually received, not what would have happened otherwise. To combat this obstacle, researchers typically rely on simulated datasets where the true ATE is known. Thus, in order for us to evaluate whether LLMs can perform causal inference, we also construct a controlled simulation setting that generates observational datasets with ground-truth ATEs. We do so using eight simulation engines that systematically vary key sources of causal difficulty, creating a more structured stress-testing framework. To ensure reproducibility, we fix the random seed for all simulations. Unless otherwise specified,  $p$ -dimensional covariates  $X$  are independently generated from a standard Normal distribution,  $\mathcal{N}(0, 1)$ . Model errors are also

independent normal  $\mathcal{N}(0, \sigma^2)$ . The weight vectors  $w$  and  $b$  are initialized using independent draws from a  $Uniform(0, 1)$  distribution.

Unless stated otherwise:

$$X \in \mathbb{R}^p, X \sim \mathcal{N}(0, I_p), \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

Each of the following engines produce  $n$  observations, including  $p$ -dimensional covariates  $X$ , treatment  $A$ , potential outcomes  $Y_0$  and  $Y_1$ , and the observed outcome  $Y$ .

## 2.1 Summary of Engines

Table 1 provides a compact overview of the specific causal challenge commonly encountered that each engine focuses on.

| Engine | Primary stressor(s)                                                 |
|--------|---------------------------------------------------------------------|
| 1      | Linear baseline; tunable confounding; constant ATE                  |
| 2      | Nonlinear outcome model; constant ATE                               |
| 3      | Interaction-heavy outcome; heterogeneous treatment effects (HTE)    |
| 4      | High-dimensional sparse confounding; HTE; subset overlap structure  |
| 5      | Weak overlap / near-positivity violations via propensity truncation |
| 6      | Nonlinear treatment assignment; nonlinear confounding               |
| 7      | Missing outcomes (MCAR and MAR)                                     |
| 8      | Covariate measurement error; heavy-tailed outcome noise             |

Table 1: Simulation engines and the causal difficulty each emphasizes.

## 2.2 Engine 1: Linear Baseline with Tunable Confounding

In any causal inference task, the most fundamental challenge is basic confounding, where covariates influence both the treatment assignment and the final outcome. The standard approach to this problem relies on simple adjustment techniques like regression or inverse probability weighting (IPW). Engine 1 is the simplest engine, with both treatment assignment and outcome linearly dependent on  $X$ , serving as the baseline case for subsequent tests to verify if the LLM can adjust for basic, well-behaved linear confounding.

For the probability of treatment assignment, we choose unit vector  $w \in \mathbb{R}^p$  and parameters  $(\alpha, \kappa)$ :

$$e(X) = \sigma(\alpha + \kappa w^\top X), A|X \sim \text{Bernoulli}(e(X)).$$

Let  $\beta \in \mathbb{R}^p$ , we define the potential outcomes as

$$Y(0) = \beta^\top X + \varepsilon, Y(1) = \beta^\top X + \tau_0 + \varepsilon.$$

In this baseline configuration, the ground truth Average Treatment Effect (ATE) is constant and equivalent to the parameter  $\tau_0$ .

The weight vectors  $w$  and  $b$  are normalized. Using Engine 1, we generate three datasets with the ground-truth ATE of  $\{0, 0.5, 1\}$ , while holding the constants  $\alpha = 1$  and  $\kappa = 1$ .

## 2.3 Engine 2: Nonlinear Outcome with Constant Effect

Real-world causal relationships are rarely perfectly linear. This engine tests whether the LLM can model these complex relationships without being explicitly prompted with structural equations. Engine 2 adopts the same treatment assignment as Engine 1, but its outcome depends nonlinearly on the first five covariates.

With treatment assignment same as Engine 1, we define the potential outcomes as

$$g(X) = 0.5X_1^2 + \sin(X_2) + 0.25X_3X_4 - 0.1X_5^3,$$

and set

$$Y(0) = g(X) + \varepsilon, Y(1) = g(X) + \tau_0 + \varepsilon$$

The ground truth ATE is  $\tau = \tau_0$ .

We generate three datasets using Engine 2, corresponding to  $\tau \in \{0, \frac{1}{2}\sqrt{2\log n}, \sqrt{2\log n}\}$ , where  $\sqrt{2\log n}$  is motivated by extreme value theory as an approximation to the expected maximum to ensure that the treatment effect is sufficiently large to not be obscured by the nonlinear component  $g(X)$ . The constants  $\alpha = 1$  and  $\kappa = 1$  are held fixed.

## 2.4 Engine 3: Interaction-Heavy Heterogeneous Treatment Effects (HTE)

In many scientific and policy settings, a treatment does not affect everyone equally, so identifying these heterogeneous treatment effects (HTE) is essential. This setting tests whether the LLM can go beyond average effects and detect effect complex covariate interactions. Engine 3 maintains the same treatment assignment as Engine 1, but its outcome model is characterized by non-linear covariate interactions as well as covariate-dependent treatment effects (HTE).

For treatment assignment, we use a (possibly different) unit vector  $v$ :

$$e(X) = \sigma(\alpha + \kappa v^\top X).$$

For outcomes, let

$$m_0(X) = 0.2X_1 + 0.2X_2 + 0.2X_3X_4, \tau(X) = \tau_0 + \tau_1X_1 + \tau_2X_2^2,$$

then

$$Y(0) = m_0(X) + \varepsilon, Y(1) = m_0(X) + \tau(X) + \varepsilon.$$

We set the ground truth Gaussian moments as if  $X \sim \mathcal{N}(0, I_p)$  then  $\mathbb{E}[X_1] = 0$  and  $\mathbb{E}[X_2^2] = 1$ , hence  $\tau = \mathbb{E}[\tau(X)] = \tau_0 + \tau_2$ .

We generate five datasets for Engine 3 that vary the strength of covariate influence on the treatment effect. The constants  $\alpha = 1$  and  $\kappa = 1$  are held fixed again.

## 2.5 Engine 4: High-Dimensional Sparse Confounding + HTE

Many modern observational datasets, like health records or genomic databases, often contain thousands of variables, but only a small fraction are true confounders. We'd like to evaluate the LLM's ability to ignore the irrelevant noise and isolate the sparse true confounders. Engine 4 defines two sparse subsets of covariates: treatment assignment depends on one subset, while the outcome is linear in a (potentially overlapping) subset. The treatment effect is specified as a function of a preselected covariate  $X_{j^*}$ .

For treatment assignment, we set

$$e(X) = \sigma(\alpha + \kappa \sum_{j \in S_A} w_j X_j)$$

For outcomes,

$$m_0(X) = \sum_{j \in S_Y} \beta_j X_j, \tau(X) = \tau_0 + \tau_1 X_{j^*},$$

$$Y(0) = m_0(X) + \varepsilon, Y(1) = m_0(X) + \tau(X) + \varepsilon.$$

With  $X_{j^*} \sim \mathcal{N}(0, 1)$ ,  $\mathbb{E}[X_{j^*}] = 0$ , so  $\tau = \tau_0$ .

For Engine 4, we generate four datasets with varying HTEs and initialize the sparse subsets to overlap at covariate  $X_1$ . The constants  $\alpha = 1$  and  $\kappa = 1$  are fixed, with  $j^* = 2$ .

## 2.6 Engine 5: Overlap/Positivity Near-Violations

A core assumption of causal inference is "positivity" or "overlap", which assumes that every individual has a non-zero chance of receiving either treatment. When this is violated, classical estimators like IPW become unstable (dividing by near-zero probabilities). We can test here how robust the LLM is to regions of data where counterfactual information is scarce. Engine 5 conducts controlled stress-testing of near-positivity violations by constraining propensity scores to lie within  $[\epsilon, 1 - \epsilon]$  and progressively shrinking  $\epsilon \downarrow 0$ .

For treatment assignment, let  $\tilde{e}(X) = \sigma(\alpha + \kappa w^\top X)$  and define

$$e_\epsilon(X) = \epsilon + (1 - 2\epsilon)\tilde{e}(X).$$

Then  $A \sim \text{Bernoulli}(e_\epsilon(X))$ .

We use Engine 5 with propensity scores  $\epsilon \in \{0.10, 0.05, 0.02, 0.01, 0.005\}$  to generate different treatment assignments with increasingly weaker overlap. Each resulting dataset is then passed through Engines 1-4 to generate outcomes using their default parameter settings as described above.

## 2.7 Engine 6: Nonlinear Treatment Assignment

In practice, treatment assignment is often complex and non-linear, and estimators that mischaracterize this assignment mechanism risk introducing severe selection bias. This setting tests the LLM's ability to adjust for confounding when the selection process itself is highly non-linear. Engine 6 specifies both treatment assignment and outcome as nonlinear functions of covariates, inducing nonlinear confounding.

For treatment assignment, define

$$s(X) = X_1 + X_2^2 - X_3X_4 + 0.5 \sin(X_5), e(X) = \sigma(\alpha + \kappa s(X)).$$

For outcomes we take one example,

$$m_0(X) = 0.5X_1 + 0.25X_2^2 - 0.2X_3X_4, Y(1) = m_0(X) + \tau_0 + \varepsilon, Y(0) = m_0(X) + \varepsilon.$$

The ground truth ATE is  $\tau = \tau_0$ .

We generate multiple datasets by varying the confounding strength parameter  $\kappa$  and the constant treatment effect  $\tau_0$ . Concretely, we fixed  $p = 20, n = 300, \alpha = 0, sd = 1$ , and a fixed random seed, and created datasets over the grid  $\tau_0 \in \{0.5, 1.0, 2.0\}$  and  $\kappa \in \{0.5, 1.0, 2.0\}$ . For each dataset we additionally record the true propensity score  $e(X)$  alongside  $(X, A, Y)$ .

## 2.8 Engine 7: Missingness (MCAR and MAR)

Observational data in the real world is almost never perfectly complete. Dealing with missingness usually requires careful data imputation, so this engine evaluates whether the LLM can maintain accurate causal estimates when the observed outcome data is heavily fragmented. Engine 7 takes a generated dataset  $(X, A, Y)$  from any complete-data engine and assigns missingness in the observed outcomes by assigning missing values either (1) completely at random via a  $\text{Bernoulli}(\pi)$  (MCAR), or (2) according to probabilities that depend on covariates and treatment (MAR).

(1) Let  $R \sim \text{Bernoulli}(\pi)$  be independent of  $(X, A, Y)$  and observe

$$Y^{\text{obs}} = \begin{cases} Y, & R = 1, \\ \text{missing}, & R = 0. \end{cases}$$

(2) Let  $\mathbb{P}(R = 1|X, A) = \sigma(\gamma_0 + \gamma_A A + \gamma^\top X)$ . Again observe  $Y$  only when  $R = 1$ .  $\tau$  is defined with respect to the *complete-data* DGP; compute analytically when applicable, otherwise via  $\hat{\tau}_{MC}$ .

Across experiments, we vary the treatment effect parameter  $\tau \in \{0, 0.5, 1.0\}$  in the underlying complete data DGP, and then apply the missingness mechanism to the resulting dataset. The ground truth ATE remains defined with respect to the complete data DGP.

## 2.9 Engine 8: Measurement Error + Heavy-Tailed Noise

Researchers often only have access to noisy proxies, with the outcome data frequently containing extreme outliers. Thus, it is imperative to examine the LLM’s ability to provide robust estimates when given imperfect, noisy real-world measurements. Engine 8 produces a dataset  $(W, A, Y)$  in which  $W$  are noisy measurements of the true covariates  $X$ . Treatment assignment  $A$  depends on covariates  $X$  in the same manner as in Engine 1, while the observed outcome  $Y$  is generated from  $W$ . Additionally, the outcome error term follows a mixture of Gaussian distributions, introducing heavy-tail noise.

Generate true  $X$  but provide observed

$$W = X + U, U \sim \mathcal{N}(0, \sigma_U^2 I_p),$$

while  $A$  and  $Y(a)$  depend on  $X$  (not  $W$ ). The dataset contains  $(W, A, Y)$ .

Replace Gaussian  $\varepsilon$  with Student- $t$ :

$$\varepsilon \sim \sigma \cdot t_v, v \in \{3, 5\},$$

or a contamination mixture:

$$\varepsilon \sim 0.95\mathcal{N}(0, \sigma^2) + 0.05\mathcal{N}(0, (10\sigma)^2).$$

If the effect is an additive constant,  $\tau = \tau_0$  still holds. Otherwise, compute via  $\hat{\tau}_{MC}$ .

Engine 8 generates four datasets with ground-truth treatment effects set to  $\{0, 2, 4, 6\}$ . Independent measurement error  $\mathcal{N}(0, 0.5^2)$  is added to the true covariates, and outcome noise follows a contamination mixture with  $w_{\text{main}} = 0.95$  weight on  $\mathcal{N}(0, 1^2)$  and 0.05 weight on  $\mathcal{N}(0, (sd_{\text{multiplier}} \cdot 1)^2)$ , where  $sd_{\text{multiplier}} = 10$ . The constants  $\alpha_0 = 1$  and  $\kappa = 1$  remain fixed.

## 3 Preliminary ICL Interface Design & Exploration

Now that we have our simulated data, we turn to how to represent it in a way that allows us to rigorously assess whether LLMs can perform causal inference by building an evaluation framework that standardizes data-to-text serialization, enforces a strict output schema, and defines accuracy and uncertainty metrics.

### 3.1 Data-to-Text Serialization

All datasets are first passed pre-cleaned to enforce consistency across engines and replicates. Columns are reordered to follow a fixed structure:  $A, Y, X_1, \dots, X_p$ . Potential outcome quantities  $Y_0$  and  $Y_1$  are removed to avoid leaking ground-truth information and to better reflect information constraints in the real-world. All variables are coerced to numeric form and rounded to three decimal points where necessary to prevent trivial cues from formatting differences, with invalid or missing entries converted to NaN. Treatment assignment  $A$  is clipped to the set  $\{0, 1\}$  to prevent formatting artifacts. This pipeline also supports row shuffling to mitigate position bias and covariate standardization to prevent scale-driven weighting errors.

In our initial design of representing data for in-context learning (ICL), for smaller samples ( $n < 200$ ), we print the entire dataset between sentinel tokens DATA\_START and DATA\_END, starting with the header line listing column names and followed by one line per observation. For larger samples (and for small samples, where we intentionally want to limit the amount of information the LLM receives), the pipeline supports a hybrid representation: it can generate a balanced subsample by drawing equal numbers of treated and untreated units without replacement. This subsample essentially preserves the texture of the data while remaining compact enough for the LLM to reliably parse. In addition, this pipeline can optionally produce summary statistics computed on the full dataset: treated and control means, standard deviations, and standardized mean differences (SMDs) for each covariate. Covariates would be ranked by their —SMDs—, with only the top imbalanced variables printed. This technique outputs the strongest confounding signals without requiring all  $p$  covariates to be displayed, especially when dimensionality is large.

### 3.2 ICL Prompt and Output Schema

To standardize model behavior and enable fully automated evaluation, we define both a fixed prompt structure and a machine-readable output schema. Each evaluation prompt follows a four-block structure (Appendix A). First, a fixed task instruction block defines the estimand and required outputs. The model is instructed to estimate the ATE  $\mathbb{E}[Y(1) - Y(0)]$  from the observational data  $(X, A, Y)$ , assuming unconfoundedness given  $X$ . The instructions require both a point estimate and a 95% confidence interval, as well as an explicit assessment of overlap. Second, optional few-shot examples may be included. Each example includes a dataset (in the same serialization format) and a correctly formatted answer. Few-shot examples must not share random seeds with evaluation instances. Third, the evaluation dataset is inserted using the serialization procedure. Depending on sample size, this may consist of the full dataset enclosed or a balanced subsample with summary statistics. Finally, the prompt concludes with an explicit reminder of the required output schema. The interval must satisfy  $\text{ci95}[0] \leq \text{ate.hat} \leq \text{ci95}[1]$ . If overlap is weak, the model should set `overlap_warning = true` and widen the interval. The method must be a concise descriptor. Outputs that are not parseable under this schema are marked as format failures and excluded from numerical scoring.

### 3.3 Stress Test Ideation

Our framework incorporates explicit "traps" designed to induce common mistakes. Trap 1 augments the dataset with an additional, post-treatment variable  $Z$  derived from the treatment and covariates, testing whether the model mistakenly adjusts for  $Z$  and measuring any resulting bias. Trap 2 includes a covariate that strongly predicts treatment assignment  $A$  but has no effect on outcome  $Y$ , assessing whether the model can distinguish treatment predictors from outcome confounders and avoid over-weighting this variable. Trap 3 utilizes Engine 4 to generate weak-overlap regimes (ie. propensities near 0 or 1), evaluating whether the model issues warnings and recommends stabilization or wider uncertainty intervals. Together, these stress tests provide a rigorous evaluation of the model's ability to perform reliable causal inferences under challenging conditions.

## 4 Refining the ICL Interface with Method-Inducing Protocols

Our preliminary ICL interface establishes a clean, standardized pipeline, from simulation to serialization to structure evaluation, but it remains generic. This level of generality proved to be both under- and over-informative. Without being guided towards a specific statistical procedure, the model may default to a convenient but suboptimal estimator, even when more appropriate causal methods exist. Simultaneously, providing the full dataset (or even a large portion of it) makes the prompt unnecessarily large, supplying the model with more information than would realistically be available in many applied settings. Thus, an effective ICL design should not depend on exposure to large quantities of information but rather only the information necessary for the intended estimation method. This next section refines the interface into a more method-aware design. We preserve the pre-cleaning pipeline but restructure the data-to-text serialization into four protocols that induce distinct causal estimation methods. Rather than presenting raw data and summary statistics, each protocol selectively reveals information required for its corresponding method, mirroring how statisticians would pull specific data views based on their chosen theoretical approach. We also extend the original prompt and output schema by incorporating protocol-specific instructions and few-shot examples, as well as expanding the required output fields to support richer diagnostics.

### 4.1 ICL Protocols for ATE Estimation

#### 4.1.1 Protocol A: Difference-in-Means

In perfectly randomized controlled trials, simple difference-in-means is the gold standard for estimating causal effects. We include this protocol to establish a baseline.

This protocol serves as a sanity check to verify that the model can reliably perform basic estimation and report uncertainty without adjustment. We split the dataset based on control and treatment groups. We then compute and return group-level summary statistics: group (treated/control), sample size ( $n$ ), mean outcome (mean  $Y$ ), and standard deviation ( $sd\ Y$ ).

#### 4.1.2 Protocol B1: Regression Adjustment from Microdata

In many small-scale studies, analysts work with raw data, so this protocol evaluates whether the LLM can process raw microdata to perform covariate adjustment.

In this protocol, we specify the maximum number of rows to return ( $j=200$ ). If the dataset contains more rows than this limit, we randomly select the required number of rows without replacement and return a raw microdata table containing the treatment A, outcome Y, and covariates X for this selected subset.

#### 4.1.3 Protocol B2: Regression Adjustment from Compressed Data

Real-world datasets often contain sensitive information, and in these cases, researchers rely on statistical summaries rather than raw records: can the model synthesize standard regression outputs to deduce the causal effect?

This protocol tests whether the model follows covariates adjustment logic and adheres to strict reporting instructions. We fit a linear regression of Y on treatment A, covariates X, and an intercept of 1, using the same microdata from B1. The model first attempts to use robust OLS with HC1 heteroskedasticity-consistent standard errors; if that fails, it falls back to manual OLS, which assumes classical homoskedasticity. The protocol returns the sample size n, the number of covariates p, the  $R^2$ , the standard error type used (HC1 or classical), and fitted coefficient of A along with its standard error.

#### 4.1.4 Protocol C: Propensity Score Stratification

When observational data contains many covariates, it is common to group individuals into "strata" based on their probability of receiving treatment. We test the LLM's ability to aggregate these stratum-specific effects into an overall ATE.

Protocol C attempts to evaluate the model's overlap sensitivity and whether it is capable of identifying positivity issues. We estimate propensity scores by fitting a logistic regression of treatment assignment on the covariates, converting this relationship into a probability score clipped between [0.001, 0.999]. We then divide the data into K bins based on the quantiles of these scores. The protocol returns bin-level summary statistics: bin weight, number of treated units, number of control units, as well as the means and standard deviations of the outcomes for both treated and control units within that bin. This provides a high-level view of how treatment and outcomes vary across comparable groups.

#### 4.1.5 Protocol D: Doubly Robust AIPW with Precomputed Nuisance Quantities

Modern causal inference relies heavily on Augmented Inverse Probability Weighting (AIPW). This protocol isolates whether the LLM can correctly compute the Doubly Robust AIPW estimator using provided nuisance components, separately from learning those components itself. Rather than applying models to the full dataset, we divide the data into training and held-out test sets. On the training set, we fit a logistic regression to estimate propensity scores and separate linear regressions on the treated and control groups to estimate the outcome functions. We then apply these fitted models to predict the values for the held-out test set. The protocol returns a per-unit table for these held-out units, containing the observed treatment A, observed outcome Y, estimated propensity score, and the predicted expected outcomes under both control and treatment

### 4.2 Guided Examples and Refined Schema

#### 4.2.1 Few-Shot Examples

We utilize few-shot examples to not only demonstrate arithmetic computations, but to teach the model how to follow the required output format and exercise appropriate statistical caution (Appendix B). Across protocols, the examples reinforce strict JSON schema adherence — explicitly reporting assumptions and structured diagnostics. They also model how to behave under edge cases, such as returning null when key quantities, like empty treatment cells or missing nuisance predictions, prevent valid estimation. In this way, the few-shot prompts train the model to behave cautiously and analytically, being transparent about assumptions and sensitive to data limitations.

### 4.2.2 Refined Output Schema

Building on our preliminary schema, the revised output schema requires a substantially richer overlap analysis, replacing a simple warning flag with a more structured assessment that supports nuanced reasoning about positivity and weak overlap (Appendix C). It also introduces a dedicated diagnostic block covering balance, missingness, and model specification risk, encouraging a more structured output rather than a short free-form note. As a result, the following revised schema strengthens statistical rigor, improves interpretability, and promotes more transparent and responsible causal reporting.

## 5 End-to-End Simulation Pipeline

Now that we have constructed eight DGPs and four method-inducing ICL protocols, we combine these components into a unified, automated testing pipeline that (i) generates synthetic datasets with known ground-truth ATEs, (ii) transforms these datasets into protocol-specific LLM inputs, and (iii) batch-feed them into the LLM to produce and evaluate causal effect estimates.

### 5.1 Large-Scale Data Generation

A single run of each engine outputs a moderately-sized dataset consisting of 300 observations and 20 covariates, along with treatment assignments and both potential and observed outcomes as defined by the engine specification. For each engine and each ground-truth  $\tau$  (as shown in Table 2), we generate 500 independent datasets, which are stored and used for all subsequent analyses corresponding to that engine and  $\tau$ .

### 5.2 From Data to LLM Prompt Construction

Each generated dataset is passed through a pre-specified protocol, converting the raw tabular format into the required text format to induce a specific estimation method (ie. group summaries, microdata, propensity score bins, or nuisance components). Once serialized, the data is inserted in a global prompt template containing strict task instructions and protocol-specific few-shot examples.

Engine 1 was additionally evaluated under a “baseline” protocol that preserves the original dataset in its raw tabular form. It is then passed through a similar set of task instructions that now explicitly guide the computation of the quantities required for AIPW estimation from the raw inputs. This baseline serves as a point of comparison for protocols designed to use structured formats to induce specific estimation strategies.

Running thousands of synchronous API calls is both time- and cost-intensive, so we utilize OpenAI’s asynchronous Batch API, submitting batches of 100 prompts five times for each engine and tau. We employ the gpt-4.1 model, as newer models currently do not support this batch processing feature. We also enforce our predefined output schema in the API request, constraining the LLM to produce exactly one properly formatted JSON object. These requests are saved to an input file and submitted to the OpenAI server. A polling function monitors job status until completion, after which the LLM’s structured responses are downloaded and stored for data extraction and evaluation.

### 5.3 Operational Considerations

Deploying ICL at scale introduces important operational constraints related to token usage, run-time, and cost. Because each dataset must be serialized into text, the total number of tokens passed to the LLM heavily depends on the chosen protocol’s data representation. Each API call is priced based on the number of input and output tokens processed, creating a tradeoff: richer data representations may better preserve information relevant for accurate causal estimation, but they come at a higher cost.

To mitigate these costs, we leverage “OpenAI’s Batch API to send asynchronous groups of requests with 50% lower costs” (OpenAI), with costs of \$1.00 per 1M input tokens and \$4.00 per 1M output tokens. We constrain model responses to a maximum of 500 tokens. Across all simulations, the global instruction portion of each prompt remains at approximately 1,000 tokens or 3,300 characters, with the baseline protocol requiring slight longer instructions due to the need to explicitly specify how to compute the required components for estimation.



The baseline protocol, which retains the full data representation, is particularly expensive: when applied to Engine 1, its data portion of the prompt produces nearly 26,000 tokens. Due to this substantial cost, we restrict the baseline evaluation to Engine 1 only. This highlights a key motivation for protocol design, compressing datasets into only the information necessary to induce a specific estimation procedure. Thus, our 4 protocols drastically reduce token usage. Most engines use Protocol D, which provides only the nuisance components required for estimation, resulting in just 1,000-1,500 tokens for the data portion and runtimes of approximately 1-1.5 hours. The primary exception is Engine 7, which incorporates missing data, so we provide both the full covariate set and the nuisance components to ensure the model has sufficient information. This increases the data prompt size to roughly 8,000 tokens and extends runtime to over 3 hours, making it one of the more expensive configurations.

In total, running all simulations across the 8 engines, including test runs, incurred a cost of approximately \$196 after tax. This reinforces that economic consideration is key in the design and evaluation of LLM-based estimation pipelines.

| Engine | Varying Parameters                          |                       |                     |                  |
|--------|---------------------------------------------|-----------------------|---------------------|------------------|
|        | $\tau$                                      | Overlap $\varepsilon$ | Conf. Str. $\kappa$ | Obs. Prob. $\pi$ |
| 1      | 0, 0.5, 1                                   |                       |                     |                  |
| 2      | 0, 1.689, 3.378                             |                       |                     |                  |
| 3      | (0,0,0), (0,1,4), (0,0,4), (1,1,1), (2,4,6) |                       |                     |                  |
| 4      | (0,0), (0,1), (1,1), (1,4)                  |                       |                     |                  |
| 5      | (0,0,0), (0,1,4), (2,4,6)                   | 0.10, 0.05, 0.005     |                     |                  |
| 6      | 0.5, 1, 2                                   |                       | 0.5, 1, 2           |                  |
| 7      | 0, 0.5, 1                                   |                       |                     | 0.5, 0.7, 0.9    |
| 8      | 0, 2, 4, 6                                  |                       |                     |                  |

Table 2: Engine Parameters (Varying)

| Engine | Constant Parameters                                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | $\alpha = 1, \kappa = 1, w, b \sim \text{Uniform}(0, 1)^{p \times 1}$                                                                        |
| 2      | $\alpha = 1, \kappa = 1, w \sim \text{Uniform}(0, 1)^{p \times 1}$                                                                           |
| 3      | $\alpha = 1, \kappa = 1, v \sim \text{Normal}(0, 1)^{p \times 1}$                                                                            |
| 4      | $\alpha = 1, \kappa = 1, w, b \sim \text{Uniform}(0, 1)^{p \times 1}, SA = [0, 1, 2], SY = [0, 3, 4], j_\star = 1$                           |
| 5      | $\alpha = 1, \kappa = 1, w \sim \text{Uniform}(0, 1)^{p \times 1}$ , and engine 3 parameters                                                 |
| 6      | $\alpha = 0$                                                                                                                                 |
| 7      | $\alpha = 1, \kappa = 1, w, b \sim \text{Uniform}(0, 1)^{p \times 1}$ , missing = "mcar"                                                     |
| 8      | $\alpha = 1, \kappa = 1, w, b \sim \text{Uniform}(0, 1)^{p \times 1}, \sigma_U = 0.5, w_{\text{main}} = 0.95, sd_{\text{multiplier}} = 10.0$ |

Table 3: Engine Parameters (Constant)

## 5.4 Evaluation Metrics

To evaluate whether the LLM behaves as a credible causal estimator, we establish standard evaluation metrics to compute across simulation replicates. For each replicate, the true treatment effect  $\tau_{\text{true}}$  is extracted from the simulation identifier, and compared to the LLM estimate  $\hat{\tau}$ . We assess point estimation performance using average raw bias, average absolute bias, and root mean squared error (RMSE) to capture systematic deviation and overall error magnitude. And to better understand the full distribution of these errors, we track their standard deviations alongside the minimum, 25th percentile, median, 75th percentile, and maximum for both raw and absolute bias. In addition, we summarize the distribution of  $\hat{\tau}$  using its sample mean, variance, and standard deviation to characterize variability across replicates.

For uncertainty quantification, we extract 95% confidence intervals (CIs) from the LLM outputs. We compute coverage probability, which is the proportion of replicates where  $\tau_{\text{true}}$  lies within the reported CI, against the nominal 0.95 level. We then further summarize the reported CIs using the

average lower and upper bounds, the average CI width, and the standard deviation of the widths. We can also use the CI width to reverse-engineer the LLM’s implied standard error and variance, enabling us to directly compare the model’s reported uncertainty and its actual empirical variability.

The resulting performance patterns vary across simulation engines. Differences in the distribution of estimated treatment effects, interval widths, and coverage behavior can be observed across settings with different data-generating mechanisms. These results are presented and visualized in detail in the plots in the following sections.

These findings motivate a direct comparison with classical causal inference estimators in the next section. By benchmarking the LLM against established methods such as regression adjustment, propensity score methods, and doubly robust estimators, we can better assess whether the observed strengths and limitations of the LLM reflect fundamental challenges in the data or shortcomings specific to the LLM-based approach.

- Identifiers / basic information
  - ‘engine’
  - ‘true\_ATE’: true average treatment effect  $\tau_{\text{true}}$
  - ‘n\_valid’: total number of valid points
- Point estimation
  - ‘average\_bias’: mean of estimation error  $(\hat{\tau} - \tau_{\text{true}})$
  - ‘sd\_bias’: standard deviation of estimation error  $(\hat{\tau} - \tau_{\text{true}})$
  - ‘bias\_min’, ‘bias\_q25’, ‘bias\_median’, ‘bias\_q75’, ‘bias\_max’: minimum, first quartile, median, third quartile, and maximum of estimation error  $(\hat{\tau} - \tau_{\text{true}})$
  - ‘average\_absolute\_bias’: mean of absolute estimation error  $|\hat{\tau} - \tau_{\text{true}}|$
  - ‘sd\_absolute\_bias’: standard deviation of absolute estimation error  $|\hat{\tau} - \tau_{\text{true}}|$
  - ‘abs\_bias\_min’, ‘abs\_bias\_q25’, ‘abs\_bias\_median’, ‘abs\_bias\_q75’, ‘abs\_bias\_max’: minimum, first quartile, median, third quartile, and maximum of absolute estimation error  $|\hat{\tau} - \tau_{\text{true}}|$
  - ‘RMSE’: root mean squared error  $\sqrt{\text{mean}((\hat{\tau} - \tau_{\text{true}})^2)}$
  - ‘median\_absolute\_error’: median of absolute estimation error  $|\hat{\tau} - \tau_{\text{true}}|$
  - ‘mean\_tau\_hat’: average of all ATE estimates  $\bar{\hat{\tau}}$
  - ‘sample\_variance\_tau\_hat’: variance of ATE estimates  $\bar{\hat{\tau}}$
  - ‘sample\_sd\_tau\_hat’: standard deviation of ATE estimates  $\bar{\hat{\tau}}$
  - ‘se\_mean\_tau\_hat’: standard error of ATE estimates  $\frac{\text{SD}_{\hat{\tau}}}{\sqrt{n_{\text{valid}}}}$
- Uncertainty
  - ‘avg\_ci\_lower’: average lower bound of the 95% confidence intervals
  - ‘avg\_ci\_upper’: average upper bound of the 95% confidence intervals
  - ‘avg\_ci\_width’: average width of the 95% confidence intervals
  - ‘sd\_ci\_width’: variability of the confidence interval widths
  - ‘avg\_se\_from\_ci’: average standard error implied by the model  $\left(\frac{U-L}{2 \times 1.96}\right)$
  - ‘avg\_var\_from\_ci’: average variance implied by the model  $\left(\frac{U-L}{2 \times 1.96}\right)^2$
  - ‘n\_ci’: total number of valid confidence intervals
  - ‘nominal level’: target statistical confidence level 95%
  - ‘coverage’: empirical coverage probability  $L \leq \tau_{\text{true}} \leq U$

## 6 Classical Propensity-Score Estimators for Benchmark Comparison

To access whether LLM behaves like a credible causal estimator based on in-context learning, we utilize several traditional benchmark methods with transparent failure modes as well as unambiguous statistical features. Using the same simulated datasets generated in Section 5.1, these classical estimators demonstrate a clear reference for evaluating LLM performance, enabling us to distinguish arithmetic errors, overlap insensitivity, or unstable uncertainty reporting. In this section, we mainly focus on propensity-score-based benchmark estimators, combining with simpler benchmark methods like naive difference-in-means and regression adjustment.

### 6.1 Simple classical estimators

We begin with two straightforward baselines that anchor the benchmark throughout the study: naive difference-in-means and regression adjustment.

#### 6.1.1 Naive difference-in-means

This approach estimates the ATE by directly comparing the average observed outcomes in the treated and control groups, without adjusting for any covariates.

$$\hat{\tau}_{DM} = \bar{Y}_1 - \bar{Y}_0$$

where  $\bar{Y}_1$  and  $\bar{Y}_0$  represent the sample mean outcomes in the treated and control groups respectively. If treated and control sample sizes are  $n_1$  and  $n_0$ , while sample variances are  $s_1^2$  and  $s_0^2$  correspondingly. Then we can compute standard error:

$$\widehat{SE}(\hat{\tau}_{DM}) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}$$

Then a 95% confidence interval is given by:

$$\hat{\tau}_{DM} \pm 1.96 \cdot \widehat{SE}(\hat{\tau}_{DM})$$

This estimator fails to consider confounding in observational data obviously, but it serves as a fundamental benchmark and sanity check, which is the first step of a thousand miles journey.

#### 6.1.2 Regression adjustment

This method estimates the ATE by regressing the outcome on treatment and observed covariates, and interpreting the coefficient on treatment as the adjusted effect estimate. We fit the linear model:

$$Y_i = \beta_0 + \beta_A A_i + \beta_X^\top X_i + \varepsilon_i$$

and the estimated ATE

$$\hat{\tau}_{RA} = \hat{\beta}_A$$

It normally relies on bootstrap confidence intervals. Unlike the no-adjustment difference-in-means method, regression adjustment approach explicates conditions on observed covariates and therefore indicates observed confounding through an outcome model.

## 6.2 Role and setup of Propensity-Score Methods

Among these diversiform classical estimators, propensity-score approach stands out for considering overlap, positivity, and instability under observational confounding. This is significantly crucial to our simulation framework since most engines are designed to stress-test weak overlap, nonlinear confounding, and high-dimensional imbalance. Therefore, propensity-score-based estimators could offer

not only average treatment effect estimates, but systematic diagnostics regarding reliability, stability as well as interpretability.

These methods measure treatment assignment mechanism through propensity score, rather than modeling the outcome directly.

$$e(X) = \Pr(A = 1 \mid X)$$

Since confounding occurs when variables affect both outcome and treatment assignment, while weak overlap represents treated and control groups aren't sufficiently comparable, calculating 1-dimensional propensity score ingeniously avoids these problems. Adopting a logistic-regression procedure provides a more reasonable and unbiased implementation. For each unit  $i$ , we estimate:

$$\hat{e}(X_i) = \text{expit}(\hat{\alpha}_0 + \hat{\alpha}^\top X_i)$$

where  $\text{expit}(t) = (1 + e^{-t})^{-1}$ . This choice provides a simple, interpretable, and reproducible treatment model that is well aligned with the benchmark's goal of comparing LLM outputs to standard causal estimators rather than to highly tuned black-box procedures. In addition, the same general estimation strategy is already used in the Protocol C construction part, where estimated propensity scores are computed from logistic regression before being clipped and converted into quantile-based bin summaries for the LLM.

Because extreme propensity score causes extreme inverse weights, leading to systematic errors on following models, we apply a truncation guardrail by replacing raw propensity score with

$$\hat{e}(X_i) \leftarrow \min\{\max(\hat{e}(X_i), \varepsilon), 1 - \varepsilon\}$$

We define  $\varepsilon$  as 0.01, 0.02 or even 0.05 so that this clipping method could prevent explosive inverse weights efficiently. However, this truncation should be interpreted as a computational guardrail rather than a solution to positivity failure. When propensity scores concentrate near 0 or 1, the underlying problem is weak overlap, and that issue must be diagnosed explicitly rather than hidden by clipping alone.

### 6.3 Diagnostics for Overlap and Stability

In our benchmark, overlap diagnostics are important as several simulation engines are designed to generate weak-overlap or near-positivity-violation regimes. Therefore, diagnostics on whether treated and control groups remain comparable reflect the reliability and stability of propensity-score-based methods. Every estimator introduced in the remainder of this section is accompanied by a common set of overlap and stability diagnostics.

Our first diagnostic layer is based on the distribution of the estimated propensity scores themselves. For each replicate, we record the propensity range and key quantiles overall and by treatment group, for example the minimum, 5th percentile, median, 95th percentile, and maximum. The summaries indicate whether the treated and control samples occupy similar regions of covariate space or whether one group is pushed toward near-deterministic treatment assignment.

It is universally acknowledged that the overlap problem is inevitable for weighting estimators, so we compute the effective sample size (ESS) separately for treated and control groups:

$$ESS_a = \frac{\left(\sum_{i:A_i=a} w_i^{(a)}\right)^2}{\sum_{i:A_i=a} \left(w_i^{(a)}\right)^2}, \quad a \in \{0, 1\}$$

Here, ESS quantifies how much information remains after reweighting. This distinction is important because a large nominal sample can still correspond to a very small effective sample if a few units dominate the weighted analysis. These quantities are treated as required diagnostics, and low ESS or extreme maximum weights are used as explicit triggers for overlap warnings in the ICL-aligned workflow.

Besides, stability check for stratification-based estimators is required. After dividing the sample into propensity-score bins, we record the number of treated and control units within each bin. Existing of empty bins shows that local treated-control comparisons cannot be formed in that part of the

propensity-score distribution, and therefore signal an explicit overlap violation rather than a mere warning.

We design these diagnostics not only to compose a complete procedure, but a significant part of how estimator behavior is evaluated. Thus, it is treated as a calibrated behavior in our evaluation metric. A good estimator should not only return a point estimate and interval, but also react sensibly when overlap deteriorates. In a nutshell, the overlap diagnostics define a common screening layer for all treatment-model-based benchmarks in this section.

## 6.4 Stratification-Based Estimation

Among the propensity-score-based benchmark estimators, stratification plays a particularly important role because it provides the clearest connection between classical causal estimation and the ICL protocols developed earlier. Stratification groups observations into regions of similar estimated treatment probability and compares treated and control outcomes within those regions. This makes the estimator both interpretable and diagnostically transparent: each component of the final ATE estimate is tied to a subset of the sample where treated and untreated units are intended to be locally comparable.

$\hat{e}(X_i)$  represents the estimated propensity score for unit  $i$ . We partition the sample into  $K$  bins based on quantiles of  $\hat{e}(X_i)$ , with  $K=5$  or  $K=10$  used as practical defaults. Let  $S(i) \in \{1, \dots, K\}$  denote the bin assignment for unit  $i$ , and let  $n_k$  be the number of observations in bin  $k$ . Within each bin, we define the within-bin contrast:

$$\hat{\tau}_k = \bar{Y}_{1k} - \bar{Y}_{0k}$$

where  $\bar{Y}_{1k}$  and  $\bar{Y}_{0k}$  are the treated and control mean outcomes. The overall stratified estimator is then formed as a weighted average of these local contrasts:

$$\hat{\tau}_{\text{Strat}} = \sum_{k=1}^K \hat{p}_k \hat{\tau}_k, \quad \hat{p}_k = \frac{n_k}{n}$$

This estimator can be interpreted as comparing treated and control outcomes only after coarse balancing on the estimated treatment mechanism, while preserving a direct and easy-to-audit decomposition into bin-level contributions.

For uncertainty quantification, our default choice is the nonparametric bootstrap, since it provides a unified approach across simulation regimes and integrates naturally with the project’s broader evaluation of interval coverage and interval width. When within-bin standard deviations are available, an analytic approximation can also be reported:

$$\widehat{\text{Var}}(\hat{\tau}_{\text{Strat}}) \approx \sum_{k=1}^K \hat{p}_k^2 \left( \frac{s_{1k}^2}{n_{1k}} + \frac{s_{0k}^2}{n_{0k}} \right)$$

with the usual normal-approximation confidence interval constructed from the resulting standard error. In practice, however, bootstrap intervals are preferable in this benchmark because they keep the uncertainty workflow comparable across stratification, weighting, and matching estimators, especially in finite-sample settings where overlap deteriorates.

## 6.5 Weighting-Based Estimation

We next consider weighting-based estimators, which use the estimated propensity score to reweight observations rather than partitioning them into bins. The goal is to construct a pseudo-population in which treatment assignment is no longer systematically related to the observed covariates, so that a weighted comparison of outcomes can be interpreted as an estimate of the ATE. Compared with stratification, weighting uses the propensity score more continuously, but it is also more sensitive to weak overlap.

### 6.5.1 Horvitz-Thompson IPW

Our weighting benchmark is the Horvitz-Thompson (HT) estimator. Define the inverse-probability weights:

$$w_i^{(1)} = \frac{A_i}{\hat{e}(X_i)}, \quad w_i^{(0)} = \frac{1 - A_i}{1 - \hat{e}(X_i)}$$

Then the HT estimator is

$$\hat{\tau}_{\text{IPW-HT}} = \frac{1}{n} \sum_{i=1}^n \left( \frac{A_i Y_i}{\hat{e}(X_i)} - \frac{(1 - A_i) Y_i}{1 - \hat{e}(X_i)} \right)$$

This estimator is useful as it's the most direct implementation of inverse probability weighting, while it's highly sensitive to weak overlap.

### 6.5.2 Hájek IPW

We also implement the normalized Hájek estimator, which improves finite-sample stability by normalizing the weighted means within treatment groups.

$$\bar{Y}_{1,w} = \frac{\sum_i A_i Y_i / \hat{e}(X_i)}{\sum_i A_i / \hat{e}(X_i)}, \quad \bar{Y}_{0,w} = \frac{\sum_i (1 - A_i) Y_i / (1 - \hat{e}(X_i))}{\sum_i (1 - A_i) / (1 - \hat{e}(X_i))}$$

Then the estimator is:

$$\hat{\tau}_{\text{IPW-H}} = \bar{Y}_{1,w} - \bar{Y}_{0,w}$$

Relative to HT estimator, this normalization reduces the influence of a small number of observations with large inverse weights and is therefore often more stable in practice.

### 6.5.3 Stabilized IPW

A further refinement is stabilized IPW, which rescales the usual inverse-probability weights by the marginal treatment prevalence. Let

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n A_i$$

The stabilized weights are

$$\tilde{w}_i^{(1)} = \frac{\hat{p} A_i}{\hat{e}(X_i)}, \quad \tilde{w}_i^{(0)} = \frac{(1 - \hat{p})(1 - A_i)}{1 - \hat{e}(X_i)}$$

These weights do not change the target effect under correct specification, but they can reduce the variance induced by highly variable raw inverse weights. In this benchmark, we treat stabilized IPW as a practical variance-reduction version of the same weighting framework.

Across all weighting estimators, overlap diagnostics remain essential. Large maximum weights, low effective sample size, and propensity scores concentrated near 0 or 1 are clear signs that the weighted comparison may be unstable. For uncertainty quantification, our default approach is the nonparametric bootstrap, which keeps interval construction comparable across weighting, stratification, and the more overlap-robust alternatives introduced next.

### 6.5.4 Augmented Inverse Probability Weighting

As a final benchmark estimator, we include augmented inverse probability weighting (AIPW), which combines propensity-score modeling with outcome regression. Unlike the PS-only estimators above, AIPW uses both the estimated treatment probability  $\hat{e}(X)$  and the estimated outcome models  $\hat{\mu}_0(X)$  and  $\hat{\mu}_1(X)$ , and is doubly robust: it remains consistent if either the propensity model is correctly specified or the outcome models are correctly specified, assuming overlap. In our benchmark, AIPW

is included as a concise classical comparator that extends the weighting framework while remaining more robust to model misspecification than single-model estimators. The estimator is given by

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left[ \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{A_i}{\hat{e}(X_i)} (Y_i - \hat{\mu}_1(X_i)) - \frac{1 - A_i}{1 - \hat{e}(X_i)} (Y_i - \hat{\mu}_0(X_i)) \right]$$

In this section, we treat AIPW simply as an additional benchmark estimator. More detailed implementation choices, such as cross-fitting, influence-function-based standard errors, and refined overlap diagnostics, are deferred to the dedicated doubly robust discussion.

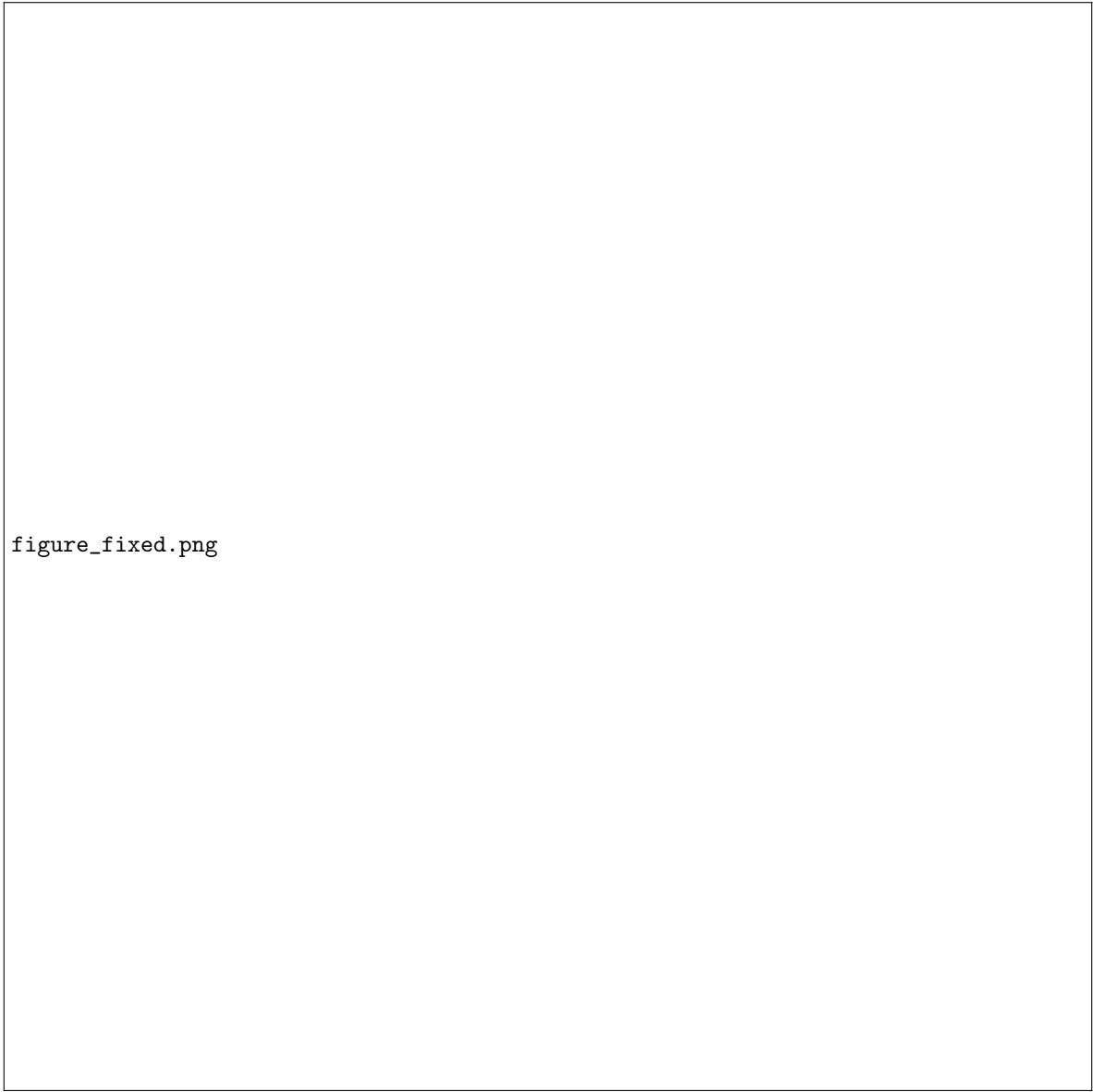
In summary, this section introduces the main classical estimators used to benchmark LLM-based causal effect estimates in our study. Beginning with difference-in-means and regression adjustment as reference baselines, and then extending to propensity-score-based methods such as stratification, inverse probability weighting, and AIPW, these estimators provide a structured set of statistical comparators on the same simulated datasets. Together, they allow us to evaluate not only whether ICL-based estimates approach the true ATE, but also whether their behavior is consistent with established causal methods under confounding, weak overlap, and other challenging regimes.

## 7 LLM vs. Classical Estimation in Simulation Setting

In this section, we evaluate the LLM’s performance, both raw-data input and ICL, against classical causal estimators in our simulation setting to assess its ability to function as a reliable causal effect estimator. We observe that (1) our method-inducing ICL prompts substantially improve estimation accuracy and uncertainty calibration compared to raw-data prompting, and (2) ICL-based LLM estimators achieve accuracy comparable to classical estimators in low causal effect settings but exhibit increasing bias and more conservative uncertainty in high causal effect settings, with notable advantages in specific nonlinear environments.

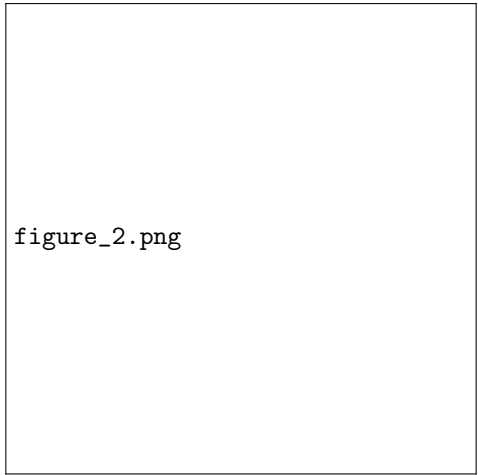
### 7.1 Raw-Data Prompting vs. Method-Inducing ICL in Simplest Setting

We start our analysis with Engine 1, as it is the simplest setting in our simulation environment, with treatment assignment and outcomes both generated from linear functions of covariates. We aim to apply ICL to more complex nonlinear, high-dimensional, or noisy situations should it perform well in a simple setting.



figure\_fixed.png

(a) [SUB-CAPTION]



figure\_2.png

(b) [SUB-CAPTION]

Figure 1: [CAPTION]



Figure 1a displays the key evaluation metrics (RMSE, absolute bias, coverage, 95% CI width) computed under each true  $\tau$  (0, 0.5, 1) from the outputs of the LLM under ICL Protocol D (LLM-D), the LLM under baseline raw data prompting (LLM-B), and each of the classical estimators (DR/L, Diff, Strat, RA). The error bands in the absolute bias and average CI width plots indicate the 95% confidence intervals for the mean metrics, computed as  $\pm 1.96$  standard errors, which were derived from the sample standard deviation of the absolute bias and CI width (respectively) and then scaled by the square root of the valid sample size. We see that unsurprisingly, difference-in-means performs poorly. Interestingly, we observe that LLM-D performs comparably to the other classical estimators and outperforms LLM-B. Protocol D, which gives the LLM estimator-ready nuisance components, yields lower RMSE and bias, along with better coverage, providing strong support that ICL with method-inducing protocols is a more accurate estimator than ICL with raw data. We’d like to consider uncertainty next, as we observe large CI widths under LLM-D and extremely small CI widths under LLM-B.

Figure 1b compares the difference between the LLM’s reported uncertainty with its empirical uncertainty. We plot its implied standard error, which was derived from its reported CI widths, against the actual standard deviation of the point estimates  $\hat{\tau}$ . The dashed  $y = x$  reference line indicates perfect calibration, where the reported uncertainty exactly matches the empirical uncertainty. All of LLM-B’s points lie below the dashed line, indicating that LLM-B behaves naively: when the LLM notices a pattern, it is overconfident in assuming it is correct, leading smaller reported error than what is computed empirically. This insight also aligns with the lower coverage and small CI width we discussed previously. Conversely, LLM-D behaves more conservatively when estimating uncertainty, systematically reporting standard errors that are larger than its true empirical uncertainty. Thus, we have strong evidence that in the simplest setting, our LLM under ICL with structured, method-inducing protocols produces both more accurate estimates and more reliable uncertainty than raw data alone.

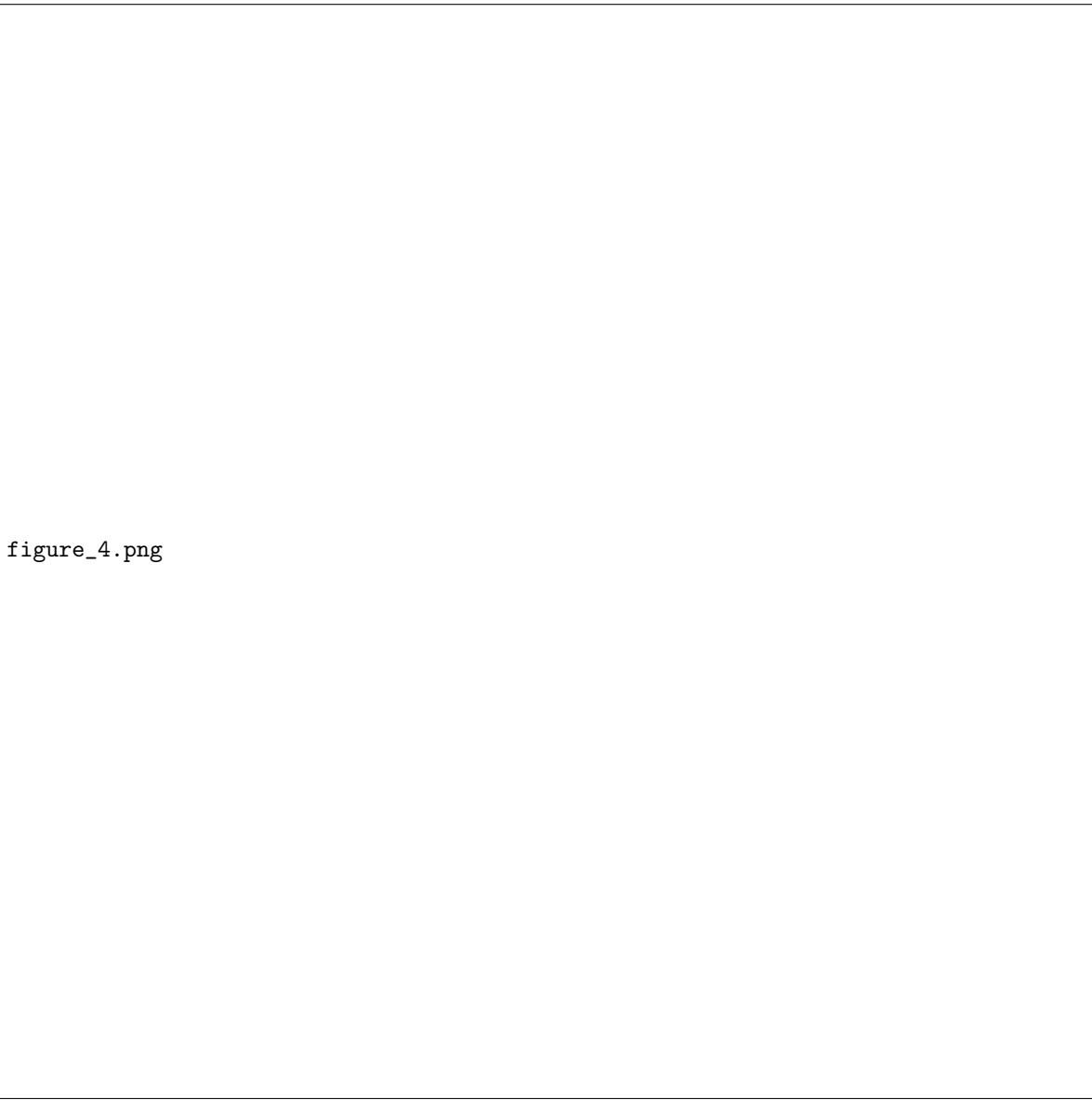
## 7.2 Analyzing Method-Inducing ICL in More Complex Settings

Now that we have seen the success of ICL in the simplest setting, we’d like further analyze its performance on the remaining more complex simulation settings.

In Figure 2a, we compare the performance of LLM-D with both the best empirical classical estimator and the best theoretical classical estimator (i.e., the one expected to perform best from a statistical standpoint). We evaluate this using the average absolute bias, with error bands again, for a fixed true  $\tau$  across all engines with that value, such that each graph corresponds to a different  $\tau$  varied over the three most common values: 0, 0.5, and 1.

While we generally do not expect our LLM to outperform the classical estimators, we see that it actually performs comparably for engines 3, 4, and 5 when there is no causal effect ( $\tau = 0$ ). But we see in Figure 2b that this behavior does not persist for the LLM as  $\tau$  increases. Larger treatment effects are associated with more negative bias, indicating that the model increasingly underestimates the true effect, which reveals a failure point of LLMs. Overall, the LLM performs comparably to classical estimators at zero or small treatment effects but exhibits increasing negative bias as  $\tau$  grows, so comparable performance is no longer guaranteed.

Most interestingly, for all values of  $\tau$  in engine 6, the LLM achieves a lower bias than both the best empirical and the best theoretical classical estimator, indicating the potential value of method-inducing LLM estimation in specific difficult regimes like nonlinear confounding. To assess whether a more flexible machine learning approach can better accommodate this setting, we evaluated an AIPW estimator utilizing 5-fold cross-fitting with a RandomForestClassifier for the propensity score and separate RandomForestRegressor outcome models (10 trees, max\_depth=5, min\_samples\_leaf=5, max\_samples=0.70). The estimator still underperforms in Engine 6, yielding a mean absolute bias of approximately 0.457 and a coverage of 20.5%. We hypothesize this is likely due to the limited per-batch sample size of 300 observations, preventing Random Forests to reliably recover the nonlinear treatment mechanisms and outcome structures, even with adding more trees. Therefore, the LLM under ICL performs competitively here, but in addition to assessing average bias, we must also consider the variability and stability of these point estimates.



figure\_4.png

Figure 3: [CAPTION]

Figure 3 addresses this by utilizing the five-number summary (minimum, 25th percentile, median, 75th percentile, and maximum) to construct box plots of the absolute bias for both LLM-D and the best empirical classical estimator, allowing us to evaluate the underlying distribution of the errors. Revisiting engines 3, 4, and 5 under no causal effect where the LLM achieved average absolute biases comparable to classical methods, we find that their interquartile ranges (IQRs) closely align, with the LLM exhibiting only slightly longer whiskers. Generally, we see that the LLM’s whiskers extend further than those of the classical estimators, indicating that when the LLM makes an error, the magnitude of that deviation is larger. However, these differences are not substantial but merely points to a mild lack of algorithmic robustness: classical estimators are constrained by strict mathematical bounds that prevent them from outputting wild numbers. Conversely, LLMs are probabilistic; while ICL Protocol D successfully trained it to be highly accurate most of the time (ie. comparable IQR boxes), it lacks hard mathematical guardrails, making it susceptible to hallucinations. The exception again is engine 6, which displays less varied error distribution in addition to lower bias.



figure\_10.png

Figure 4: [CAPTION]

Figure 4 investigates the empirical coverage and average CI widths of the LLM under Protocol D compared to the best-performing classical estimator. Under no causal effect ( $\tau = 0.0$ ), the classical estimators behave as we theoretically expect, clustering around the nominal 95% coverage threshold. The LLM exhibits over-coverage, with values frequently approaching 1.0. The bottom plots show that the LLM achieves this high coverage by generating substantially wider confidence intervals: for most engines, the LLM’s average CI width is roughly double to triple that of the classical methods. Interestingly, Engine 5 at  $\tau = 0.0$  serves as a rare exception where the LLM’s CI width match the classical estimator, resulting in comparable coverages. However, the most intriguing case emerges again in engine 6 under non-zero treatment effects ( $\tau = 0.5$  and  $1.0$ ): the classical estimator’s coverage plummets to approximately 30% while the LLM’s coverage remains at or above 95%. We can see this is still partially due to a larger average CI width, but in the scenarios with complex, non-linear confounding, the LLM’s conservative CI generation provides a safety net that our traditional estimation methods lack.

Overall, across most engines, LLM-D achieves accuracy comparable to classical estimators but tends to produce more conservative uncertainty estimates that result in wider CIs and higher (often over-) coverage. Most notably, in engine 6, which simulates a more complex nonlinear setting, LLM-D not only attains lower bias but also maintains reliable coverage where classical estimators fail, highlighting

its advantage in more challenging environments.

## 8 PSID Data

### 8.1 Data Source

The real-world analysis in this study is based on data from the Panel Study of Income Dynamics (PSID), a nationally representative longitudinal survey conducted by the University of Michigan. The PSID has tracked individuals and families in the United States since 1968, collecting detailed information on socioeconomic characteristics such as income, employment, education, and health. Due to its longitudinal structure and rich set of variables, the PSID is widely used in empirical research for studying causal relationships in observational data.

The data used in this project were obtained from the official PSID data center (<https://simba.isr.umich.edu/data/data.aspx>), where customized datasets can be generated by selecting variables across different survey waves. We extracted relevant variables from selected years to construct a dataset suitable for causal inference analysis, ensuring that the variables of interest were consistently defined and available across the chosen waves.

Here, a 'wave' refers to one round of PSID data collection. Since PSID is a panel dataset, different waves allow us to observe the same individual or family at different points in time. This is important for our design because the covariates, treatment, and outcome should not be measured at the same time. Instead, the dataset is organized in a clear time order: baseline covariates are taken from 2011, treatment is measured in 2013, and the income outcome is measured in the 2023 wave.

More specifically, the 2011 wave is used to describe each individual's pre-treatment background. These variables include demographic information, education, employment status, household structure, economic resources, and health conditions. The 2013 wave is used to define the treatment variable, which captures participation in job training, vocational training, or related human-capital investment. The 2023 wave provides the outcome variable, measured as labor income in calendar year 2022. This ordering helps reduce reverse-causality concerns because the covariates are observed before treatment and the outcome is observed after treatment. The PSID panel structure also makes it possible to link information across years. The individual file provides identifiers such as the 1968 interview number and person number, whose combination uniquely identifies individuals in the dataset. Family interview numbers, sequence numbers, and relationship-to-head variables are also used to determine each person's family unit and household role in a given wave. These identifiers are used to align observations across the 2011, 2013, and 2023 waves.

### 8.2 Variable Construction

To construct the dataset for causal analysis, we define a treatment variable, an outcome variable, and a set of covariates based on the PSID data.

The outcome variable, denoted as  $Y$ , is based on the reference person's labor income reported in the 2023 PSID wave. More specifically, the variable measures labor income in calendar year 2022. This income measure includes wages and salaries, bonuses, overtime, tips, commissions, professional practice or trade income, additional job income, and miscellaneous labor income. It excludes farm income and the labor portion of unincorporated business income. Since income is highly skewed, we use a log transformation,  $\log(\text{income} + 1)$ , to reduce the influence of very large income values and make the outcome more stable for estimation.

The treatment variable, denoted as  $A$ , is constructed from the 2013 wave. It is intended to capture participation in human-capital-enhancing activities before the outcome period. In the PSID codebook, the relevant variables include whether the head or wife/spouse attended job training programs or courses, and whether they received other degrees or certificates through a vocational school, training school, or apprenticeship program. We combine these related variables into one treatment indicator. An individual is classified as treated ( $A = 1$ ) if there is evidence of job training, vocational training, or related certification. An individual is classified as untreated ( $A = 0$ ) otherwise.

The covariates, denoted as  $X$ , are measured mainly from the 2011 wave, before the treatment is observed. In this project, these covariates describe the baseline characteristics of the 2011 Head or Reference Person and the family unit. This structure is used because the later outcome is also

measured for the Reference Person. These variables are included because they may affect both the probability of receiving training and later labor income.

The first group of covariates includes basic demographic characteristics. Age is based on the age of the 2011 Head, and sex is based on the sex of the 2011 Head. These variables account for differences in labor-market experience, earnings, and training opportunities across age and gender.

The second group measures education before treatment. Education is measured using the completed education of the Head or the corresponding individual-level completed education variable when the analytic unit is the individual. This variable captures baseline human capital, since people with different education levels may have different access to training and different future income paths.

The third group measures employment status and work history before treatment. Employment status records whether the person was working, unemployed, retired, disabled, keeping house, a student, or in another status in 2011. Weeks employed in the previous year measures how many weeks the Head worked in 2010. Current-employer experience is also included when available. Together, these variables describe the person’s baseline labor-market attachment before the treatment year.

The fourth group describes household and family structure in 2011. This includes marital status, number of children in the family unit, household size, and relationship-to-head information. For example, the number of children variable counts persons under age 18 in the family unit who are not the Head or Wife/“Wife”. These variables capture household responsibilities and family composition, which may shape both training decisions and later labor supply.

The fifth group measures baseline economic resources and financial conditions. This includes home ownership or rental status, baseline labor income, total family income, wealth, credit-card debt, student loans, and other household debt indicators. These variables describe the financial position of the household before treatment. They are included because households with different economic resources may have different access to training and different income trajectories.

The sixth group captures public assistance and economic hardship. This includes food stamps, free or reduced-cost school lunches, free or reduced-cost breakfasts, elderly meal assistance, and Social Security income receipt. These variables serve as baseline indicators of economic disadvantage and household need before treatment.

The seventh group captures baseline health conditions. This includes the Head’s general health status, hypertension, diabetes, arthritis, psychiatric or emotional problems, psychological distress, poor health among other family members, and health insurance coverage. Health is included because it may affect both the ability to participate in training and the ability to work and earn income later.

Table 4: Summary of PSID Variable Construction

| Role           | Wave | Variable group                 | Examples of variables                                                                                                                    |
|----------------|------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Outcome $Y$    | 2023 | Labor income                   | Reference person’s 2022 labor income, transformed as $\log(\text{income} + 1)$ .                                                         |
| Treatment $A$  | 2013 | Training and certification     | Job training programs/courses; vocational-school, training-school, or apprenticeship certificates.                                       |
| Covariates $X$ | 2011 | Demographics                   | Age of Head; sex of Head.                                                                                                                |
| Covariates $X$ | 2011 | Education                      | Completed education of Head; corresponding individual-level completed education measure.                                                 |
| Covariates $X$ | 2011 | Employment and work history    | Employment status; weeks employed in 2010; current-employer experience.                                                                  |
| Covariates $X$ | 2011 | Family structure               | Marital status; number of children under 18 in the family unit; household size; relationship to Head.                                    |
| Covariates $X$ | 2011 | Economic resources             | Home ownership or rental status; labor income; family income; wealth; credit-card debt; student loans; other debt indicators.            |
| Covariates $X$ | 2011 | Public assistance and hardship | Food stamps; free or reduced-cost school meals; elderly meal assistance; Social Security receipt.                                        |
| Covariates $X$ | 2011 | Health conditions              | General health; hypertension; diabetes; arthritis; psychiatric or emotional problems; psychological distress; health insurance coverage. |

### 8.3 Data Processing

The PSID data are processed to make the variables consistent across waves and suitable for causal analysis. We first link individuals across the 2011, 2013, and 2023 waves using the available PSID identifiers. The goal is to keep the same analytic unit over time so that the 2011 covariates, 2013 treatment, and 2023 outcome refer to the same individual or reference person as closely as possible. We then remove observations with missing values in either the treatment variable  $A$  or the outcome variable  $Y$ , since these two variables are required for estimating the treatment effect. For the outcome, observations with missing or invalid labor income are not used in the final estimation sample.

For the covariates  $X$ , missing values are addressed using Multiple Imputation by Chained Equations (MICE). This method fills in missing covariate values by iteratively modeling each variable using the other observed variables. The purpose is to preserve sample size while keeping the relationships among covariates as much as possible. Categorical variables are recoded into numerical formats before estimation. Binary variables are encoded as 0–1 indicators. Multi-category variables, such as employment status, marital status, and housing tenure, are either encoded numerically or expanded into dummy variables when appropriate. Continuous variables are cleaned and transformed when needed. For example, income is transformed using  $\log(\text{income} + 1)$ .

After preprocessing, the final analytic dataset is organized at the individual level and structured as  $(Y, A, X)$ . This structure is compatible with both the classical causal inference estimators and the LLM-based estimation framework used in this study. Overall, the cleaned dataset preserves the intended temporal ordering of the design: baseline characteristics from 2011, treatment from 2013, and labor income outcome from the 2023 wave.

## 9 LLM vs. Classical Estimation on PSID Data

After evaluating the LLM in controlled simulation settings, we apply the same method-inducing in-context learning framework to the PSID observational dataset. This real-data application differs from the simulation analysis in one important way: the true treatment effect is unknown. Therefore, we cannot evaluate bias, RMSE, or coverage against a known ground truth. Instead, we evaluate the PSID results through three empirical checks: whether the data provide sufficient overlap for treated-control comparisons, whether classical estimators suggest a meaningful long-run treatment effect, and whether the LLM protocol outputs are broadly consistent with the classical benchmarks.

The empirical question is whether participation in job training, vocational training, or related certification in 2013 affects long-run labor income in 2023. The outcome is measured as  $\log(\text{income} + 1)$ , so the estimated treatment effects are on the log-income scale. The treatment is measured in 2013, and the covariates are measured before treatment, mainly in 2011. This time ordering helps reduce reverse-causality concerns, although the analysis still relies on standard observational assumptions, especially unconfoundedness conditional on observed covariates and sufficient overlap between treated and control units.

Throughout this PSID analysis, we use a 5% significance level. We treat an estimate as statistically significant if its 95% confidence interval does not include zero. If the 95% confidence interval includes zero, we interpret the estimate as not statistically significant. This rule is used consistently for both the classical estimators and the LLM protocol outputs.

### 9.1 Overlap Diagnostics

A causal interpretation of the PSID estimates depends not only on model choice but also on whether treated and control units are comparable. In the final cleaned sample, there are 6,096 observations in total, including 4,240 treated units and 1,856 control units. Thus, the treated group is substantially larger than the control group, although the control group is still large enough for estimation.

To assess overlap more directly, we examine treated and control counts across propensity-score bins. The propensity score is the estimated probability of treatment conditional on observed covariates. Good overlap means that both treated and control units appear throughout the range of propensity scores. If one group dominates in certain regions, comparisons become less reliable there.

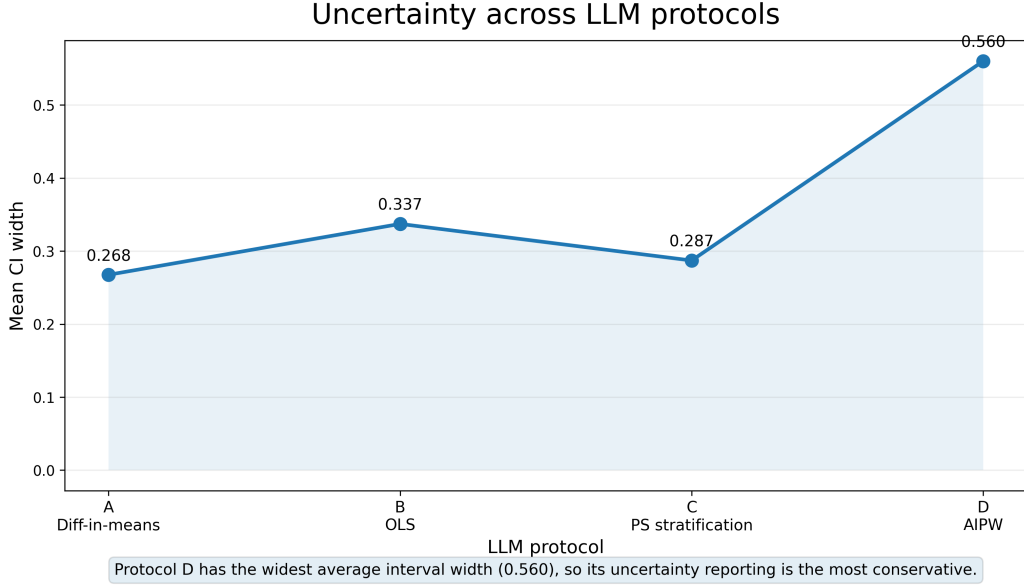


Figure 5: Overlap across propensity-score bins on the PSID data.

Figure 5 shows that overlap is not completely absent, but it is clearly weaker in the upper part of the propensity-score distribution. In the lowest bin, control units dominate. As the propensity score increases, treated units become increasingly dominant. In the bin  $(0.856, 0.922]$ , there are 1,117 treated units but only 103 control units. In the highest bin  $(0.922, 0.990]$ , there are 1,157 treated units but only 61 control units. This pattern indicates that for individuals who are highly likely to receive training, there are relatively few comparable untreated observations. As a result, adjusted estimates in these regions are less stable and should be interpreted with caution.

## 9.2 Classical Estimation Results

We first compute four classical estimators on the cleaned PSID sample: difference-in-means, OLS regression adjustment, propensity-score stratification, and AIPW (doubly robust estimation). These estimators provide a benchmark for interpreting the LLM outputs.

Table 5: Classical Causal Estimates on PSID Data

| Estimator            | ATE     | SE     | 95% CI Lower | 95% CI Upper |
|----------------------|---------|--------|--------------|--------------|
| Difference-in-means  | -0.0047 | 0.0302 | -0.0640      | 0.0545       |
| OLS adjustment       | 0.0159  | 0.0370 | -0.0566      | 0.0883       |
| PS stratification    | 0.0390  | 0.0482 | -0.0556      | 0.1336       |
| AIPW / Doubly robust | 0.0339  | 0.0438 | -0.0520      | 0.1197       |

Table 5 shows that all four classical estimators produce relatively small treatment effects. The naive difference-in-means estimate is essentially zero. The covariate-adjusted methods, including OLS, propensity-score stratification, and AIPW, are slightly positive, but in all cases the 95% confidence intervals include zero. Therefore, under the 5% significance criterion defined above, the classical analysis does not provide statistically significant evidence that 2013 job training increases 2023 log labor income.

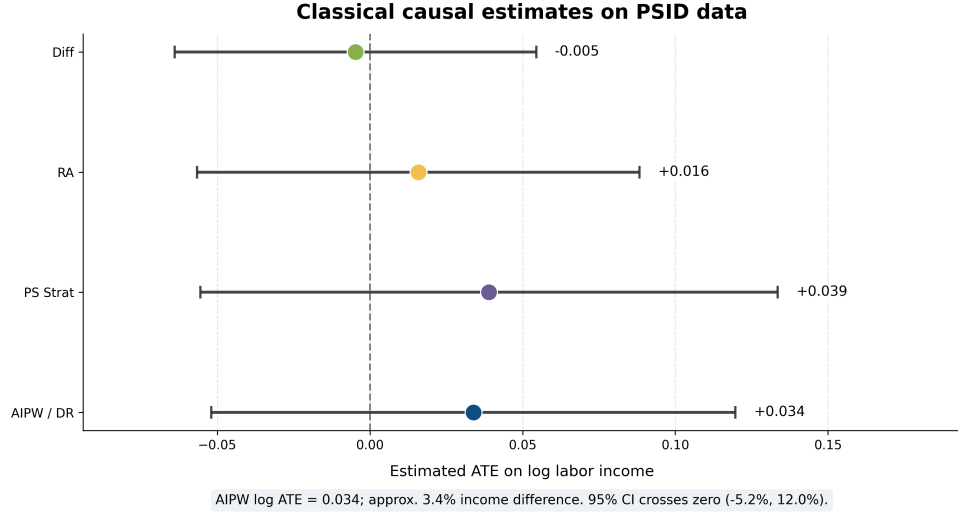


Figure 6: Classical causal estimates on PSID data. Each point is an estimated ATE and each horizontal line is the corresponding 95% confidence interval. The dashed vertical line marks zero. All confidence intervals cross zero, so none of the classical estimators provides statistically significant evidence of a long-run income effect.

Figure 6 visualizes the same result. The AIPW estimate is 0.0339, which can be loosely interpreted as about a 3.4% difference in income on the log scale. However, its confidence interval ranges from approximately  $-0.052$  to  $0.120$ , so the estimate is not statistically significant. Overall, the classical estimators suggest that any average treatment effect in this sample is small and uncertain.

### 9.3 LLM Protocol Outputs

We next compare the classical benchmarks with the LLM-based protocol outputs. For the PSID application, we focus on the run-5 setting, where each protocol is repeated five times. Protocol A corresponds to difference-in-means, Protocol B to regression adjustment, Protocol C to propensity-score stratification, and Protocol D to AIPW-style reasoning.

Table 6: Summary of LLM Protocol Outputs on PSID Data (run 5)

| Protocol                 | Mean ATE | SD ATE | Min ATE | Max ATE | Mean CI Width |
|--------------------------|----------|--------|---------|---------|---------------|
| A: Difference-in-means   | -0.031   | 0.030  | -0.056  | 0.021   | 0.268         |
| B: Regression adjustment | 0.060    | 0.071  | -0.013  | 0.168   | 0.337         |
| C: PS stratification     | -0.016   | 0.118  | -0.144  | 0.170   | 0.287         |
| D: AIPW                  | -0.012   | 0.090  | -0.091  | 0.099   | 0.560         |

The mean LLM estimates are also fairly close to zero. Protocol B gives the largest positive mean estimate, while Protocols A, C, and D are slightly negative on average. However, the protocol-level means are still small in magnitude. More importantly, the repeated runs reveal noticeable variability, especially for Protocols C and D. This means that the LLM does not return one perfectly stable estimate, but its outputs remain centered near zero rather than indicating a clear strong positive effect.



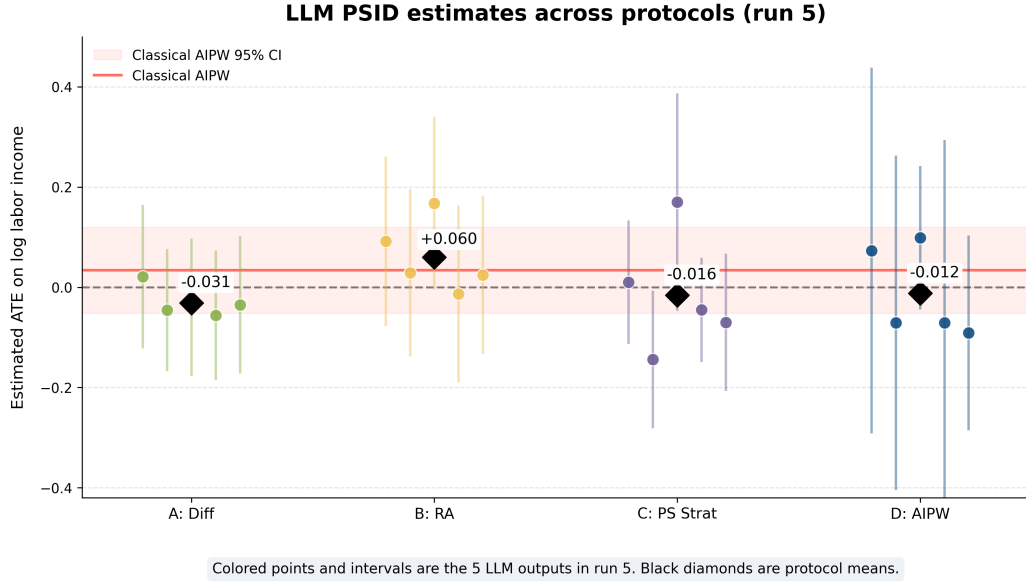


Figure 7: LLM protocol outputs on PSID data. Colored points and vertical lines show the five repeated LLM outputs and their 95% confidence intervals within each protocol. Black diamonds show the protocol-level mean ATE. The purple horizontal line and shaded band represent the classical AIPW benchmark and its 95% confidence interval.

Figure 7 shows how the LLM estimates vary across repeated runs. The LLM outputs are not expected to exactly match the classical estimators, because the LLM uses compressed, protocol-specific inputs rather than re-running the full estimation code on the full cleaned dataset. For example, Protocol A is based on group summary statistics, Protocol B is based on regression information, Protocol C is based on propensity-score strata, and Protocol D is based on precomputed nuisance quantities. Therefore, small numerical differences between LLM and classical estimates are expected.

The more important finding is that the LLM outputs are broadly consistent with the classical analysis in their substantive conclusion. The protocol-level mean estimates stay close to zero, and they do not provide stable evidence of a strong positive effect. Although some individual LLM runs produce positive estimates and others produce negative estimates, the overall pattern remains statistically uncertain.

## 9.4 LLM Uncertainty Across Protocols

We also examine how much uncertainty the LLM reports under different protocols. This connects the PSID analysis back to the simulation results, where Protocol D often produced more conservative confidence intervals.

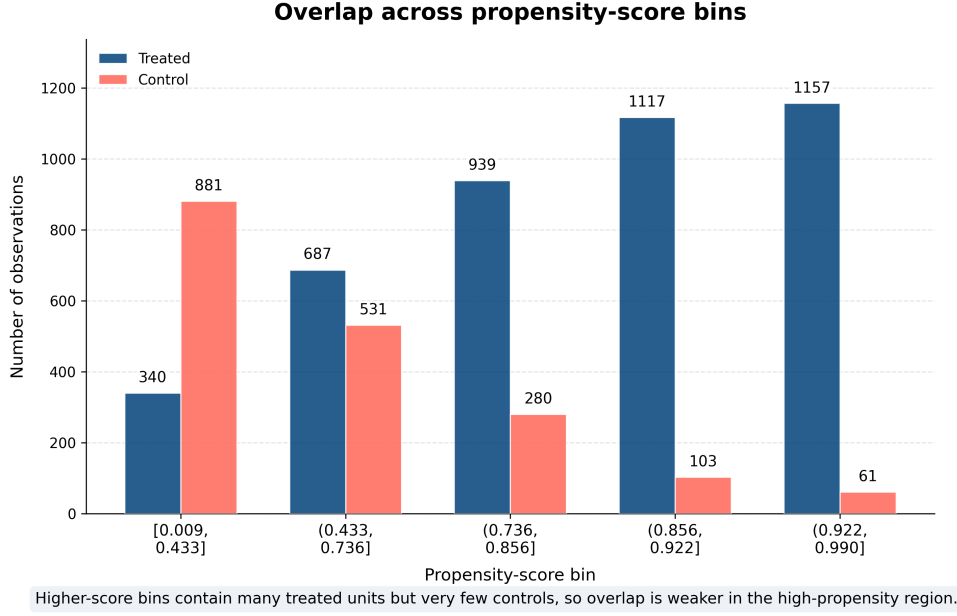


Figure 8: Mean confidence interval width across LLM protocols using run-5 outputs.

Figure 8 shows the average 95% confidence interval width across the four LLM protocols. Protocol A has an average CI width of 0.268, Protocol B has an average CI width of 0.337, Protocol C has an average CI width of 0.287, and Protocol D has an average CI width of 0.560. Protocol D therefore produces the widest intervals among the four protocols.

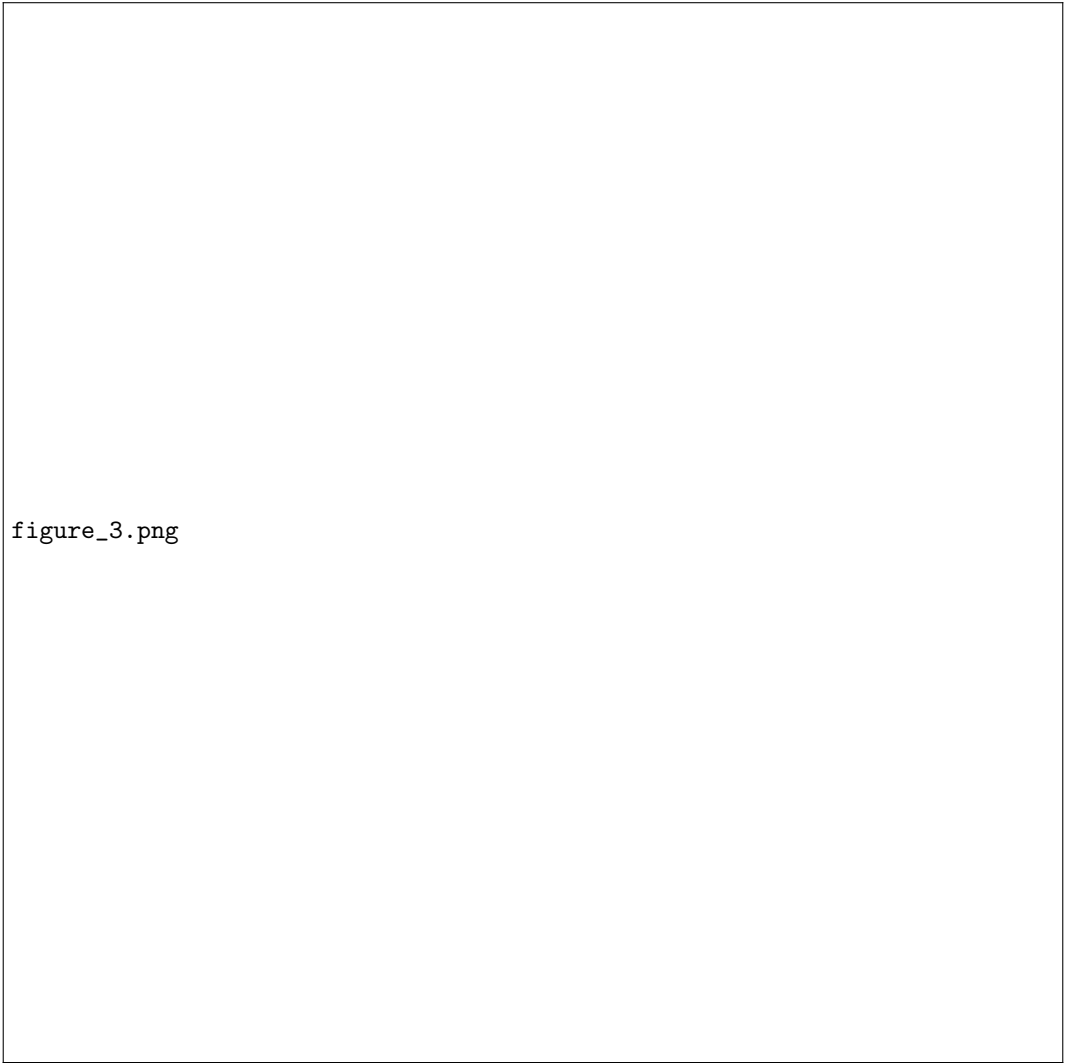
This pattern is useful because it suggests that the AIPW-style prompt does not lead the LLM to make an overconfident causal claim. Instead, Protocol D reports wider uncertainty, which is reasonable in an observational setting where the treatment is not randomly assigned and the job-training measure combines different types of training and certification experiences. In this sense, Protocol D behaves more cautiously than the other protocols.

## 9.5 Main Findings

Taken together, the PSID results support a cautious conclusion. Using the 5% significance level defined above, the classical estimators do not show statistically significant evidence that 2013 job training has a clear long-run effect on 2023 log labor income. The LLM protocol outputs, although variable across repeated runs, are broadly consistent with this finding. Their protocol means remain close to zero, and the AIPW-style protocol reports especially wide intervals, indicating conservative uncertainty quantification.

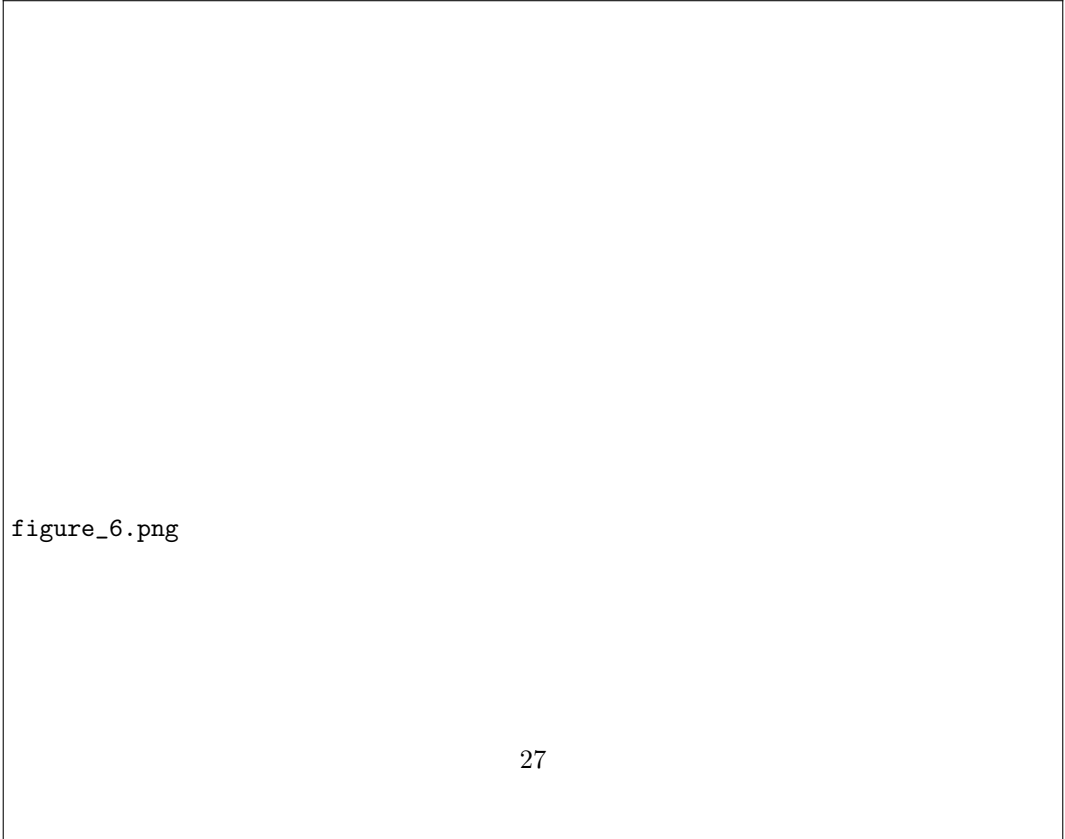
This does not prove that job training has no effect. Instead, it means that under the current treatment definition, covariate set, sample construction, and estimation framework, the data do not provide strong evidence of a nonzero average treatment effect. One likely reason is that the treatment variable aggregates multiple forms of job training and certification, which may differ substantially in quality, duration, and labor-market relevance. Any positive effects for specific subgroups may therefore be diluted in the full-sample average treatment effect.

More broadly, the PSID application serves as a useful real-world check on the simulation findings. In the simulations, method-inducing prompts helped align the LLM with classical estimator logic. In the observational PSID setting, the LLM again does not produce a strong unsupported causal claim. Instead, its outputs remain cautious and broadly in line with the classical estimators. This suggests that method-inducing in-context learning can help structure LLM-based causal reasoning, but final interpretation still depends on the same core identification conditions as in standard causal inference, especially unconfoundedness and overlap.



figure\_3.png

(a) [SUB-CAPTION]



figure\_6.png

# Appendix

## A Preliminary ICL Prompt Specification

PROMPT START

TASK\_INSTRUCTIONS

You are given observational data with covariates  $X$ , treatment  $A$  (0/1), and outcome  $Y$ . Estimate the average treatment effect (ATE) from observational data ( $X$ ,  $A$ ,  $Y$ ). Assume unconfoundedness given  $X$  and discuss overlap/positivity. Provide a point estimate and a 95% interval, with warnings if overlap is weak.

END\_TASK\_INSTRUCTIONS

serialized\_data.strip

OUTPUT\_SCHEMA

Return exactly ONE JSON object with these required fields:

```
{
  "ate_hat": <number>,
  "ci95": [<lower>, <upper>],
  "method": "<short label>",
  "assumptions": ["unconfoundedness_given_X", "overlap"],
  "overlap_warning": <true/false>,
  "notes": "<brief justification, <= 120 words>"
}
```

Rules:

The interval must satisfy  $ci95[0] \leq ate\_hat \leq ci95[1]$ .

If overlap is weak (as suggested by propensity quantiles/bins or extreme imbalance), the model should set `overlap_warning = true` and widen the interval.

Method must be a concise descriptor (e.g., "AIPW: logistic PS + linear outcome", "IPW(trimmed)", "outcome regression").

END\_OUTPUT\_SCHEMA

PROMPT END

## B Protocol D Few-Shot Examples

### Example 1 (Well-posed case)

EXAMPLE\_EVALUATION\_DATASET\_START

A Y e\_hat mu0\_hat mu1\_hat

1 3.0 0.6 1.0 2.5

0 1.2 0.4 1.0 2.0

1 2.8 0.7 1.2 2.6

0 1.1 0.5 1.0 2.1

EXAMPLE\_EVALUATION\_DATASET\_END

```
{
  "ate_hat": 1.275,
  "se_hat": 0.248,
  "ci_95": [0.789, 1.761],
  "estimator_name": "aipw_dr",
  "assumptions": ["ignorability", "overlap", "SUTVA"],
  "overlap_assessment": {
```

```

    "status": "ok",
    "e_min": 0.4,
    "e_max": 0.7,
    "notes": "Propensity scores are comfortably bounded away from 0 and 1."
  },
  "diagnostics": {
    "balance_summary": "not_provided",
    "missingness": "none (dropped_rows=0, valid_rows=4)",
    "model_specification_risk": "depends_on_nuisance_quality"
  },
  "refusal_or_caution": "ATE, SE, and CI computed using AIPW influence-function estimator."
}

```

## Example 2 (Weak overlap)

EXAMPLE\_EVALUATION\_DATASET\_START

A Y e\_hat mu0\_hat mu1\_hat

1 2.000 0.990 1.200 1.800

0 1.000 0.020 0.900 1.400

1 3.000 0.800 1.500 2.600

EXAMPLE\_EVALUATION\_DATASET\_END

```

{
  "ate_hat": 0.9333,
  "se_hat": 0.8500,
  "ci_95": [-0.7320, 2.5986],
  "estimator_name": "aipw_dr",
  "assumptions": ["ignorability", "overlap", "SUTVA"],
  "overlap_assessment": {
    "status": "warning",
    "e_min": 0.02,
    "e_max": 0.99,
    "notes": "Weak overlap: extreme propensity scores may destabilize weights."
  },
  "diagnostics": {
    "balance_summary": "not_provided",
    "missingness": "not_provided",
    "model_specification_risk": "high_due_to_extreme_weights"
  },
  "refusal_or_caution": "Inference unstable due to weak overlap; consider trimming or re-specification."
}

```

## Example 3 (Missing nuisance predictions)

EXAMPLE\_EVALUATION\_DATASET\_START

A Y e\_hat mu0\_hat mu1\_hat

1 2.000 0.600 null 1.900

0 1.000 0.400 0.800 null

EXAMPLE\_EVALUATION\_DATASET\_END

```

{
  "ate_hat": null,
  "se_hat": null,
  "ci_95": [null, null],
  "estimator_name": "aipw_dr",
  "assumptions": ["ignorability", "overlap", "SUTVA"],
  "overlap_assessment": {

```

```

    "status": "ok",
    "e_min": 0.4,
    "e_max": 0.6,
    "notes": "Cannot compute AIPW influence function due to missing nuisance predictions."
  },
  "diagnostics": {
    "balance_summary": "not_provided",
    "missingness": "Dropped rows with missing mu0_hat/mu1_hat (no valid rows).",
    "model_specification_risk": "unknown"
  },
  "refusal_or_caution": "Cannot compute ATE because required nuisance functions are missing."
}

```

## C Revised Output Schema

```

STRUCTURED_SCHEMA = {
  "type": "json_schema",
  "name": "module2_output",
  "schema": {
    "type": "object",
    "properties": {
      "ate_hat": {
        "type": ["number", "null"]
      },
      "se_hat": {
        "type": ["number", "null"]
      },
      "ci_95": {
        "type": "array",
        "items": {
          "type": ["number", "null"]
        },
        "minItems": 2,
        "maxItems": 2
      },
      "estimator_name": {
        "type": "string"
      },
      "assumptions": {
        "type": "array",
        "items": {"type": "string"}
      },
      "overlap_assessment": {
        "type": "object",
        "properties": {
          "status": {
            "type": "string",
            "enum": ["ok", "warning", "violation", "unknown"]
          },
          "e_min": {
            "type": ["number", "null"]
          },
          "e_max": {
            "type": ["number", "null"]
          }
        }
      }
    }
  }
}

```

```

        "notes": {
            "type": "string"
        }
    },
    "required": ["status", "e_min", "e_max", "notes"],
    "additionalProperties": False
},
"diagnostics": {
    "type": "object",
    "properties": {
        "balance_summary": {
            "type": ["string", "null"]
        },
        "missingness": {
            "type": ["string", "null"]
        },
        "model_specification_risk": {
            "type": ["string", "null"]
        }
    },
    "required": [
        "balance_summary",
        "missingness",
        "model_specification_risk"
    ],
    "additionalProperties": False
},
"refusal_or_caution": {
    "type": ["string", "null"]
}
},
"required": [
    "ate_hat",
    "se_hat",
    "ci_95",
    "estimator_name",
    "assumptions",
    "overlap_assessment",
    "diagnostics",
    "refusal_or_caution"
],
"additionalProperties": False
}
}

```